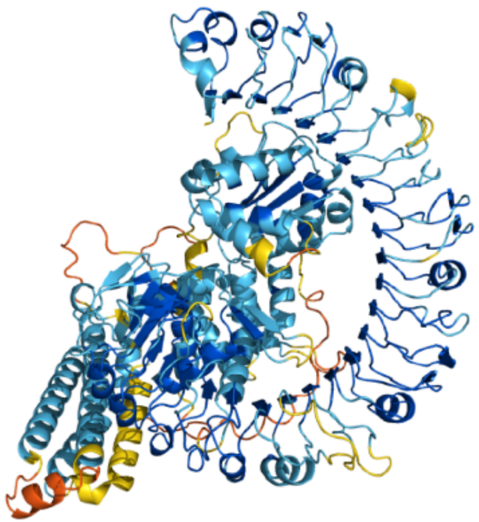
Neural networks and symmetry

- Recent years have seen neural networks and deep learning play an increasingly important role both in **science** and in **broader society.**



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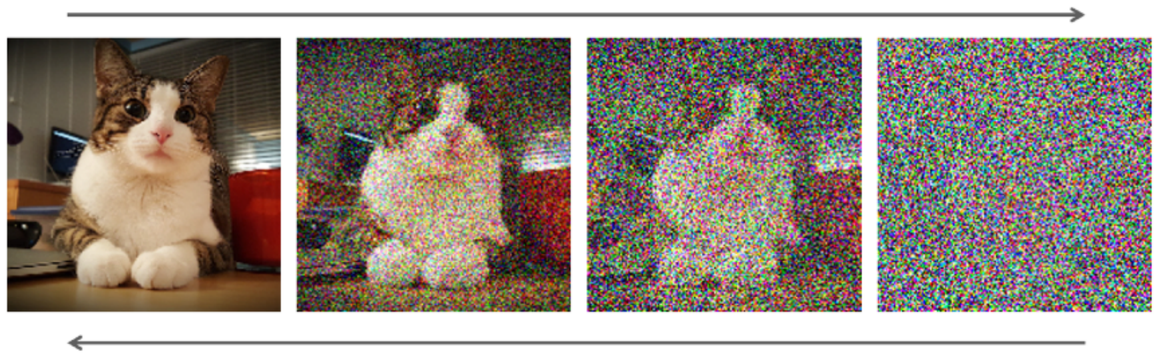

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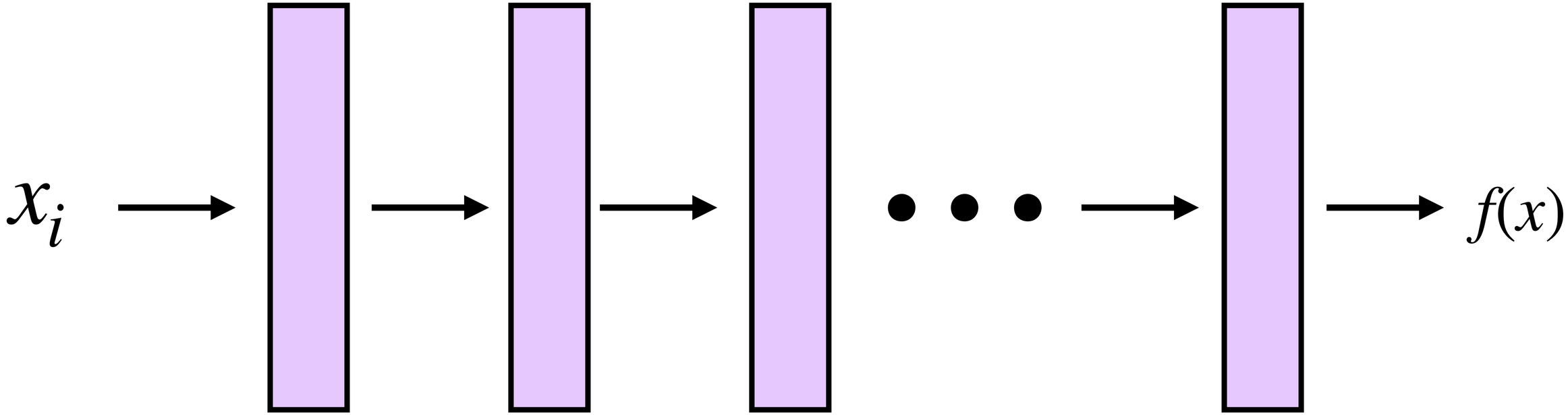
- This talk will be about **general principles** to organize the construction and understanding of neural networks. One such principle is that of **symmetry**.

What is a neural network?

- A neural network is a way to parametrize a **function** of many variables in a computer in a **useful manner** in terms of many numbers: $W_{ij}^{(n)}, b_i^{(n)}$



$$x_i^{(1)} = \tanh(W_{ij}^{(1)} x_j + b_i^{(1)}) \qquad x_i^{(n)} = \tanh(W_{ij}^{(n)} x_j^{(n-1)} + b_i^{(n)})$$